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Outboard motor and drive shaft assembling method

Abstract

An outboard motor includes a case having a gear chamber, a drive shaft arrangement hole, and a propeller shaft arrangement hole, a drive shaft, a propeller shaft, a drive gear, a driven gear, a gear position setting member provided in front of the driven gear in the case and configured to set a position of the driven gear in a front-rear direction, and a screw mechanism configured to move the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member. The screw mechanism includes a first screw provided in the case and a second screw provided on the gear position setting member and configured to be screwed with the first screw.

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Patent No.	Application Date	Country	CPC
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The disclosure of Japanese Patent Application No. 2021-209506 filed on Dec. 23, 2021, including specification, drawings and claims is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to an outboard motor and a drive shaft assembling method for assembling a drive shaft integrally formed with a drive gear to a lower case of an outboard motor.

BACKGROUND ART

(3) Many outboard motors include a power source provided at an upper portion of an outboard motor, a drive shaft extending downward from the power source, a propeller shaft provided at a lower portion of the outboard motor and extending in a front-rear direction, a propeller attached to the propeller shaft, and a gear mechanism provided at the lower portion of the outboard motor and configured to transmit rotation of the drive shaft to the propeller shaft.

(4) In such an outboard motor, the gear mechanism and the like are disposed in a lower case provided at the lower portion of the outboard motor. Specifically, a gear chamber is provided in the lower case, and the gear mechanism is accommodated in the gear chamber. The lower case is provided with a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and the drive shaft is disposed in the drive shaft arrangement hole. The lower case is provided with a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber, and the propeller shaft is disposed in the propeller shaft arrangement hole.

(5) The gear mechanism includes a drive gear coupled to a lower end portion of the drive shaft and a driven gear coupled to a front end portion of the propeller shaft. To transmit the rotation of the drive shaft extending in an upper-lower direction to the propeller shaft extending in the front-rear direction, bevel gears are used as the drive gear and the driven gear. The gear mechanism of many outboard motors includes two driven gears. That is, the gear mechanism of a contra-rotating propeller outboard motor includes two driven gears to transmit the rotation of the drive shaft to each of two propeller shafts. Also in an outboard motor including one propeller shaft, the gear mechanism is provided with two driven gears in many cases to switch the direction of the rotation transmitted from the drive shaft to the propeller shaft by a clutch. In both the gear mechanism of the contra-rotating propeller outboard motor and the gear mechanism of the outboard motor including one propeller shaft, one of the two driven gears is disposed in front of the drive gear and meshes with the drive gear from front of the drive gear, and the other driven gear is disposed in rear of the drive gear and meshes with the drive gear from rear of the drive gear.
(6) In the contra-rotating propeller outboard motor having such a configuration in the related art, the drive shaft and the drive gear are separately formed, and the drive gear is fitted to the lower end portion of the drive shaft. The propeller shaft and the driven gear are also separately formed, and the driven gear is fitted to the front end portion of the propeller shaft. In addition, in the outboard motor having the above-described configuration of including one propeller shaft in the related art, the drive shaft and the drive gear are separately formed, and the drive gear is fitted to the lower end portion of the drive shaft. In the outboard motor including one propeller shaft, each driven gear is coupled to the propeller shaft by a clutch. Due to this structure, the driven gear and the propeller shaft are separately formed.
(7) A contra-rotating propeller outboard motor (**1**) shown in FIG. 2 of Patent Literature 1 below includes a pinion gear (**18**) and a lower drive shaft (**172**). The pinion gear (**18**) corresponds to a drive gear, and the lower drive shaft (**172**) corresponds to a drive shaft. In the outboard motor (**1**), the lower drive shaft (**172**) and the pinion gear (**18**) are formed separately, and the pinion gear (**18**) is fitted to the lower drive shaft (**172**). The outboard motor (**1**) further includes a front gear (**21**), a rear gear (**22**), an inner shaft (**231**), and an outer shaft (**232**). The front gear (**21**) and the rear gear (**22**) correspond to driven gears, and the inner shaft (**231**) and the outer shaft (**232**) correspond to propeller shafts. In the outboard motor (**1**), the inner shaft (**231**) and the front gear (**21**) are formed separately, and the front gear (**21**) is fitted to the inner shaft (**231**). The outer shaft (**232**) and the rear gear (**22**) are separately formed, and the rear gear (**22**) is fitted to the outer shaft (**232**). The above-described reference numerals in parentheses are reference numerals described in Patent Literature 1.
(8) Patent Literature 1: JP2016-94116A
(9) In the outboard motors in the related art, the drive shaft and the drive gear are separately formed, and the drive gear is fitted to the lower end portion of the drive shaft. However, when the torque of the power source is increased, it may be difficult to stabilize the transmission of the power of the power source from the drive shaft to the propeller shaft with a structure in which the drive gear is fitted to the drive shaft. Therefore, it is conceivable to integrally form the drive gear with the drive shaft. By integrally forming the drive gear with the drive shaft, the coupling strength between the drive shaft and the drive gear can be increased, and the stability of power transmission from the drive shaft to the propeller shaft can be increased.
(10) However, when the drive gear is integrally formed with the drive shaft, it may be difficult to assemble the drive shaft and the drive gear into the lower case.
(11) For example, in the above-described contra-rotating propeller outboard motor in the related art, the operation of assembling the drive shaft, the two propeller shafts, the drive gear, and the two driven gears into the lower case is generally performed as follows. First, of the two driven gears, the driven gear disposed in front of the drive gear (hereinafter referred to as the “front driven gear”) is inserted into the gear chamber through the propeller shaft arrangement hole from rear of the

lower case, and is attached to the inside of the gear chamber. Next, the drive gear is inserted into the gear chamber through the propeller shaft arrangement hole from rear of the lower case, and the drive shaft is inserted through the drive shaft arrangement hole from above the lower case. In the gear chamber, the drive gear is fitted to the lower end portion of the drive shaft while being meshed with the front driven gear, and the drive shaft is attached to the drive shaft arrangement hole. Next, the driven gear disposed in rear of the drive gear and the two propeller shafts are inserted into the propeller shaft arrangement hole from rear of the lower case, and are assembled into the gear chamber or the propeller shaft arrangement hole.
(12) When the drive gear is integrally formed with the drive shaft, the drive shaft integrally formed with the drive gear needs to be assembled into the lower case. As a method for assembling the drive shaft integrally formed with the drive gear into the lower case, the following two methods are conceivable.
(13) As shown in FIG. **10**A, the first method is a method in which a drive shaft **122** integrally formed with a drive gear **121** is inserted into a drive shaft arrangement hole **126** from above a lower case **123**, the drive gear **121** is inserted into a gear chamber **125**, and the drive shaft **122** is attached to the inside of the drive shaft arrangement hole **126**.
(14) In the second method, as shown in FIG. **10**B, the drive shaft **122** integrally formed with the drive gear **121** is inserted into the gear chamber **125** from rear of the lower case **123** through a propeller shaft arrangement hole **124**, and the drive shaft **122** is inserted into the drive shaft arrangement hole **126** from below and attached to the inside of the drive shaft arrangement hole **126**.
(15) However, it is difficult to adopt the first method for the following reason. That is, as can be seen from FIG. **10**A, the drive shaft arrangement hole **126** has a diameter smaller than a diameter of the drive gear **121**, and thus the drive gear **121** cannot be inserted into the drive shaft arrangement hole **126**. Therefore, by the first method, the drive shaft **122** integrally formed with the drive gear **121** cannot be assembled into the lower case **123**. In this regard, increasing the diameter of the drive shaft arrangement hole **126** to be larger than the diameter of the drive gear **121** allows the drive shaft **122** integrally formed with the drive gear **121** to be assembled into the lower case **123**. However, when the diameter of the drive shaft arrangement hole **126** is larger than the diameter of the drive gear **121**, the lateral width of the lower case **123** increases and the resistance to water during navigation increases. Therefore, it is not preferable to increase the diameter of the drive shaft arrangement hole **126** to be larger than the diameter of the drive gear **121**.
(16) In addition, it is difficult to adopt the second method for the following reason. That is, as shown in FIG. **10**B, a front driven gear **127** needs to be attached to the inside of the gear chamber **125** before the drive shaft **122** integrally formed with the drive gear **121** is inserted into the propeller shaft arrangement hole **124**. In a state in which the front driven gear **127** is attached to the inside of the gear chamber **125**, the front driven gear **127** becomes an obstacle, and the drive shaft **122** integrally formed with the drive gear **121** cannot be inserted into the drive shaft arrangement hole **126** from below.
(17) The present disclosure is made in view of, for example, the above-described problems, and an object of the present disclosure is to provide an outboard motor and a drive shaft assembling method in which a drive shaft and a drive gear can be assembled into a lower case even when the drive gear is integrally formed with the drive shaft.
SUMMARY
(18) In order to solve the above problem, there is provided an outboard motor including: a case having a gear chamber, a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber; a drive shaft disposed in the drive shaft arrangement hole and extending in an upper-lower direction; a propeller shaft disposed in the propeller shaft arrangement hole and extending in a front-rear direction; a drive gear disposed in the gear chamber

and coupled to a lower end side of the drive shaft, the drive gear being a bevel gear; a driven gear disposed in front of the drive gear in the gear chamber, coupled to a front end side of the propeller shaft, and configured to mesh with the drive gear, the driven gear being a bevel gear; a gear position setting member provided in front of the driven gear in the case and configured to set a position of the driven gear in the front-rear direction; and a screw mechanism configured to move the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member, the axis of the gear position setting member extending in the front-rear direction. The screw mechanism includes a first screw provided in the case and having an axis extending in the front-rear direction and a second screw provided on the gear position setting member, having an axis extending in the front-rear direction, and configured to be screwed with the first screw.
(19) In order to solve the above problem, there is provided a drive shaft assembling method for an outboard motor, the outboard motor including: a case having a gear chamber, a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber; a drive shaft disposed in the drive shaft arrangement hole and extending in an upper-lower direction; a propeller shaft disposed in the propeller shaft arrangement hole and extending in a front-rear direction; a drive gear disposed in the gear chamber and integrally formed with a lower end side of the drive shaft, the drive gear being a bevel gear; a driven gear disposed in front of the drive gear in the gear chamber, coupled to a front end side of the propeller shaft, and configured to mesh with the drive gear, the driven gear being a bevel gear; a gear position setting member provided in front of the driven gear in the case and configured to set a position of the driven gear in the front-rear direction; and a screw mechanism configured to move the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member that extends in the front-rear direction, the drive shaft assembling method for assembling the drive shaft integrally formed with the drive gear into the case, and the drive shaft assembling method including: inserting the drive shaft into the drive shaft insertion hole from the gear chamber after inserting the drive shaft into the gear chamber through the propeller shaft insertion hole in a state where the driven gear is disposed in front of a set position where the driven gear is meshed with the drive gear; inserting a tool that rotates the gear position setting member into the gear chamber through the propeller shaft insertion hole and attaching the tool to the gear position setting member; and rotating the gear position setting member by the tool to move the driven gear to the set position to mesh with the drive gear.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an overall view showing an outboard motor according to an embodiment of the present disclosure;
(2) FIG. **2** is a sectional view showing the inside of a lower portion of the outboard motor according to the embodiment of the present disclosure;
(3) FIG. **3** is a sectional view showing an enlarged upper gear mechanism, an enlarged clutch, and the like in FIG. **2**;
(4) FIG. **4** is a sectional view showing an enlarged lower gear mechanism and the like in FIG. **2**;
(5) FIG. **5** is a sectional view showing a lower gear chamber in the outboard motor according to the embodiment of the present disclosure;
(6) FIG. **6** is a schematic view showing a gear position setting member in the outboard motor according to the embodiment of the present disclosure as viewed from front;
(7) FIG. **7**A is a schematic view showing the gear position setting member in FIG. **6** as viewed

from left;
(8) FIG. **7**B is a schematic view showing a tool and a cross section of the gear position setting member taken along a cutting line VII-VII in FIG. **6**;
(9) FIGS. **8**A to **8**D are schematic views showing a drive shaft assembling operation process according to the embodiment of the present disclosure;
(10) FIGS. **9**A to **9**C are schematic views showing another embodiment of a screw mechanism, a tool attachment portion, and the like of the outboard motor of the present disclosure; and
(11) FIGS. **10**A and **10**B are schematic views showing difficulties in assembling a drive shaft integrally formed with a drive gear into a lower case in an outboard motor in the related art.
DESCRIPTION OF EMBODIMENTS
(12) An outboard motor according to an embodiment of the present disclosure includes: a case having a gear chamber, a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber; a drive shaft disposed in the drive shaft arrangement hole and extending in an upper-lower direction; and a propeller shaft disposed in the propeller shaft arrangement hole and extending in a front-rear direction. The outboard motor according to the embodiment of the present disclosure further includes a drive gear and a driven gear. The drive gear is disposed in the gear chamber and coupled to a lower end side of the drive shaft. The drive gear is a bevel gear. The driven gear is disposed in front of the drive gear in the gear chamber and coupled to a front end side of the propeller shaft. The driven gear is a bevel gear and meshes with the drive gear. The outboard motor according to the embodiment of the present disclosure further includes a gear position setting member and a screw mechanism. The gear position setting member is a member that is provided in front of the driven gear in the case and that sets a position of the driven gear in the front-rear direction. The screw mechanism is a mechanism that moves the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member that extends in the front-rear direction. The screw mechanism includes a first screw provided in the case and having an axis extending in the front-rear direction, and a second screw provided on the gear position setting member, having an axis extending in the front-rear direction, and screwed with the first screw.
(13) According to the outboard motor of the present embodiment, the drive shaft can be assembled into the case even when the drive gear is integrally formed with the drive shaft. That is, by rotating the gear position setting member about the axis and moving the gear position setting member forward, the driven gear disposed in the gear chamber can be moved forward. When the driven gear moves forward, the driven gear moves away from the opening of the drive shaft arrangement hole into the gear chamber that communicates with the inside of the gear chamber. As a result, a vacant region below the drive shaft arrangement hole expands in the gear chamber. Therefore, after the driven gear is moved forward, the drive shaft integrally formed with the drive gear is inserted into the gear chamber from rear of the case through the propeller shaft arrangement hole, and the drive shaft is inserted into the drive shaft arrangement hole from below and attached to the inside of the drive shaft arrangement hole. Thereafter, the gear position setting member is rotated in an opposite direction, the driven gear is moved rearward together with the gear position setting member, so that the driven gear can be meshed with the drive gear.
(14) A drive shaft assembling method according to the embodiment of the present disclosure is a method in which, in the outboard motor according to the embodiment of the present disclosure including the drive gear integrally formed with the drive shaft on the lower end side of the drive shaft, the drive shaft integrally formed with the drive gear is assembled into the case. The drive shaft assembling method according to the embodiment of the present disclosure includes: a drive shaft inserting step of inserting the drive shaft into the drive shaft insertion hole from the gear chamber after inserting the drive shaft into the gear chamber though the propeller shaft insertion hole in a state where the driven gear is disposed in front of a set position where the driven gear is

meshed with the drive gear; a tool attaching step of inserting a tool that rotates the gear position setting member into the gear chamber through the propeller shaft insertion hole and attaching the tool to the gear position setting member; and a driven gear moving step of rotating the gear position setting member by the tool to move the driven gear to the set position to mesh with the drive gear. According to the drive shaft assembling method of the present embodiment, the drive shaft integrally formed with the drive gear can be easily assembled into the case.

EMBODIMENTS

(15) An outboard motor according to an embodiment of the present disclosure will be described. Upper (Ud), lower (Dd), front (Fd), rear (Bd), left (Ld), and right (Rd) directions described in the embodiment follow arrows drawn on lower right of FIGS. **1** to **8**D.

Outboard Motor

(16) FIG. **1** shows an outboard motor **1** according to the embodiment of the present disclosure. As shown in FIG. **1**, the outboard motor **1** is a contra-rotating propeller outboard motor. The outboard motor **1** includes an engine **2** that is a power source, a front propeller **3**, an outer propeller shaft **4** to which the front propeller **3** is attached, a rear propeller **5**, an inner propeller shaft **6** to which the rear propeller **5** is attached, and a power transmission mechanism **7** that transmits power of the engine **2** to the propeller shafts **4**, **6**. The engine **2** is disposed at an upper portion of the outboard motor **1**, the propeller shafts **4**, **6** are disposed at a lower portion of the outboard motor **1**, and the power transmission mechanism **7** is disposed between the engine **2** and the propeller shafts **4**, **6**. The power transmission mechanism **7** includes an upper drive shaft **21**, an upper gear mechanism **22**, a clutch **27**, an intermediate drive shaft **35**, a coupling member **36**, a lower drive shaft **38**, and a lower gear mechanism **45**.

(17) The engine **2** is covered with a bottom cowl **8** and a top cowl **9**. The upper drive shaft **21** is disposed in an upper case **10**. The upper gear mechanism **22**, the clutch **27**, the intermediate drive shaft **35**, the coupling member **36**, the lower drive shaft **38**, and the lower gear mechanism **45** are disposed in a lower case **11** provided at the lower portion of the outboard motor **1**. Front end portions of the outer propeller shaft **4** and the inner propeller shaft **6** are disposed in the lower case **11**. The lower case **11** is a specific example of a “case”, and the lower drive shaft **38** is a specific example of a “drive shaft”. The outer propeller shaft **4** and the inner propeller shaft **6** are specific examples of a “propeller shaft”.

Lower Case

(18) FIG. **2** shows a cross section as viewed from left, which is obtained by cutting components provided at the lower portion of the outboard motor **1** along a plane including axes of the drive shafts **21**, **35**, **38** and the propeller shafts **4**, **6** and extending in the front-rear direction and the upper-lower direction. In FIG. **2**, the lower case **11** is formed in a substantially box shape having an opened upper portion, and is provided with, at the upper portion of the lower case **11**, a lid member **12** that covers a part of the upper portion of the lower case **11**. The lid member **12** is formed with a drive shaft insertion hole **13**.

(19) The lower case **11** is provided with an upper gear chamber **14** at a front upper portion therein. The drive shaft insertion hole **13** communicates with the upper gear chamber **14**. The lower case **11** is provided with a lower gear chamber **15** at a front lower portion. The lower case **11** is formed with a drive shaft arrangement hole **16** between the upper gear chamber **14** and the lower gear chamber **15**, which connects the upper gear chamber **14** and the lower gear chamber **15**. The drive shaft arrangement hole **16** extends in the upper-lower direction and communicates with the lower gear chamber **15** from above the lower gear chamber **15**. The lower case **11** is formed with a propeller shaft arrangement hole **17** in rear of the lower gear chamber **15**. The propeller shaft arrangement hole **17** extends in the front-rear direction and communicates with the lower gear chamber **15** from rear of the lower gear chamber **15**. The lower gear chamber **15** is a specific example of a “gear chamber”.

Upper Drive Shaft

(20) As shown in FIG. **1**, the upper drive shaft **21** extends in the upper-lower direction, and an upper end of the upper drive shaft **21** is connected to the engine **2**. As shown in FIG. **2**, a lower end of the upper drive shaft **21** is inserted into the drive shaft insertion hole **13**. The upper drive shaft **21** rotates in one direction by the power of the engine **2**. The rotation direction of the upper drive shaft **21** is defined as a positive direction.

Upper Gear Mechanism and Clutch

(21) FIG. **3** is an enlarged view of the upper gear mechanism **22**, the clutch **27**, and the like in FIG. **2**. The upper gear mechanism **22** is a mechanism that generates rotation in a reverse direction relative to the rotation of the upper drive shaft **21**. The upper gear mechanism **22** is disposed in the upper gear chamber **14**. As shown in FIG. **3**, the upper gear mechanism **22** includes a reverse drive gear **23**, a reverse intermediate gear **24**, and a reverse output gear **25**.

(22) The reverse drive gear **23**, which is a bevel gear, is disposed at an upper portion of the upper gear chamber **14** such that a portion formed with teeth faces downward, and is rotatably supported to the lid member **12** by a bearing. The reverse drive gear **23** is, for example, splined (fitted) to a lower end portion of the upper drive shaft **21**, and rotates integrally with the upper drive shaft **21**.

(23) The reverse intermediate gear **24** is a bevel gear and is disposed at a front portion of the upper gear chamber **14** such that a portion formed with teeth faces rearward. The reverse intermediate gear **24** is integrally formed with a shaft, and the shaft of the reverse intermediate gear **24** is rotatably supported to the lower case **11** by a bearing.

(24) The reverse output gear **25**, which is a bevel gear, is disposed at a lower portion of the upper gear chamber **14** such that a portion formed with teeth faces upward, and is rotatably supported to the lower case **11** by a bearing. The reverse output gear **25** is formed with a through hole **26** in a central portion thereof. An upper end portion of the intermediate drive shaft **35** is inserted into the through hole **26** and is moved away from the through hole **26**.

(25) The reverse drive gear **23** meshes with the reverse intermediate gear **24**, and the reverse intermediate gear **24** meshes with the reverse output gear **25**. When the reverse drive gear **23** rotates in the positive direction, the rotation is transmitted to the reverse output gear **25** through the reverse intermediate gear **24**, and the reverse output gear **25** rotates in a reverse direction.

(26) The clutch **27**, which is a dog clutch, has a cylindrical shape, and is formed with a clutch pawl **28** on each of an upper end surface and a lower end surface. The clutch **27** is formed with a groove **29** over the entire circumference of an outer peripheral surface thereof. The clutch **27** is disposed between the reverse drive gear **23** and the reverse output gear **25**. In the upper gear chamber **14**, the upper end portion of the intermediate drive shaft **35** penetrates the through hole **26** of the reverse output gear **25** and enters between the reverse drive gear **23** and the reverse output gear **25**. The clutch **27** is coupled to the upper end portion of the intermediate drive shaft **35** in a manner of being not movable in a circumferential direction and movable in an axial direction of the intermediate drive shaft **35**. Accordingly, the clutch **27** and the intermediate drive shaft **35** rotate integrally, and the clutch **27** can move in the upper-lower direction relative to the intermediate drive shaft **35**.

(27) The lower case **11** is provided with a shift member **30** and a clutch control unit **31**. One end of the shift member **30** is inserted into the groove **29** of the clutch **27** so that the clutch **27** is rotatable relative to the shift member **30**. The other end of the shift member **30** is connected to an actuator (not shown) provided in the clutch control unit **31**. When the actuator is operated, the shift member **30** can be moved in the upper-lower direction, and the clutch **27** can be moved in the upper-lower direction. When the clutch **27** moves upward, the clutch pawl **28** on the upper end surface of the clutch **27** engages with a clutch pawl **32** on a lower end surface of the reverse drive gear **23**. Accordingly, the rotation of the upper drive shaft **21** is directly transmitted to the intermediate drive shaft **35**, and the intermediate drive shaft **35** rotates in the positive direction. On the other hand, when the clutch **27** moves downward, the clutch claw **28** on the lower end surface of the clutch **27** engages with a clutch claw **33** on an upper end surface of the reverse output gear **25**. Accordingly,

the rotation of the reverse output gear **25** is transmitted to the intermediate drive shaft **35**, and the intermediate drive shaft **35** rotates in the reverse direction.
(28) (Intermediate Drive Shaft and Coupling Member)
(29) The intermediate drive shaft **35** extends in the upper-lower direction, and is located below the upper drive shaft **21** and is coaxial with the upper drive shaft **21**. The upper end portion of the intermediate drive shaft **35** is coupled to the clutch **27** as described above. A lower end portion of the intermediate drive shaft **35** is coupled to the lower drive shaft **38** by a coupling member **36**. The coupling member **36** has a tubular shape, and is formed with a spline on an inner peripheral surface thereof. The intermediate drive shaft **35** is also formed with a spline on an outer peripheral surface of a lower end portion thereof. The lower drive shaft **38** is also formed with a spline on an outer peripheral surface of an upper end portion thereof. The lower end portion of the intermediate drive shaft **35** is inserted into an upper portion of the coupling member **36** and is splined (fitted) to the coupling member **36**. The upper end portion of the lower drive shaft **38** is inserted into a lower portion of the coupling member **36** and is splined (fitted) to the coupling member **36**. The coupling member **36** is rotatably supported to the lower case **11** by a bearing **37**.
(30) (Lower Drive Shaft)
(31) FIG. **4** is an enlarged view of the lower drive shaft **38**, the lower gear mechanism **45**, and the like in FIG. **2**. The lower drive shaft **38** extends in the upper-lower direction, and is located below the intermediate drive shaft **35** and is coaxial with the intermediate drive shaft **35**. As shown in FIG. **4**, the lower drive shaft **38** is disposed in the drive shaft arrangement hole **16**, and is rotatably supported to the drive shaft arrangement hole **16** by bearings **39**, **40**. A spacer **41** that sets the arrangement of inner rings of the bearings **39**, **40** and a nut **42** that fixes the inner rings of the bearings **39**, **40** to the lower drive shaft **38** are attached to the lower drive shaft **38**.
(32) The lower drive shaft **38** is integrally formed with a main drive gear **46** at a lower end portion thereof. The lower drive shaft **38** and the main drive gear **46** are integrally formed into one component, and are formed by, for example, forging and cutting.
(33) (Lower Gear Mechanism and Propeller Shaft)
(34) The lower gear mechanism **45** is a mechanism that transmits the power of the engine **2** transmitted through the upper drive shaft **21**, the upper gear mechanism **22**, the clutch **27**, the intermediate drive shaft **35**, the coupling member **36**, and the lower drive shaft **38** to the outer propeller shaft **4** and the inner propeller shaft **6**. The lower gear mechanism **45** is disposed in the lower gear chamber **15**. As shown in FIG. **4**, the lower gear mechanism **45** includes the main drive gear **46**, a front driven gear **47**, and a rear driven gear **49**. Although the main drive gear **46** is integrally formed with the lower drive shaft **38**, the main drive gear **46** is one of components of the lower gear mechanism **45** in terms of function. The main drive gear **46** is a specific example of a “drive gear”, and the front driven gear **47** is a specific example of a “driven gear”.
(35) The main drive gear **46** is a bevel gear, and is disposed in an upper portion of the lower gear chamber **15** such that a portion formed with teeth faces downward.
(36) The front driven gear **47** is a bevel gear and is disposed in a front portion of the lower gear chamber **15** such that a portion formed with teeth faces rearward. The front driven gear **47** is rotatably supported to the lower case **11** by a bearing **48**. The front driven gear **47** is disposed in front of the main drive gear **46** and meshes with the main drive gear **46**.
(37) The rear driven gear **49** is a bevel gear, and is disposed in a rear portion of the lower gear chamber **15** such that a portion formed with teeth faces forward. The rear driven gear **49** is rotatably supported to the lower case **11** by a bearing **50**. The rear driven gear **49** is disposed in rear of the main drive gear **46** and meshes with the main drive gear **46**.
(38) As shown in FIG. **2**, the outer propeller shaft **4** is formed in a tubular shape and extends in the front-rear direction. A front end portion of the outer propeller shaft **4** is disposed in the propeller shaft arrangement hole **17**, and the outer propeller shaft **4** is rotatably supported to the lower case **11** by bearings **51**, **52**. The rear driven gear **49** is splined (fitted) to the front end portion of the outer

propeller shaft **4**. Accordingly, the rear driven gear **49** and the outer propeller shaft **4** rotate integrally. The front propeller **3** is attached to a rear end portion of the outer propeller shaft **4**.
(39) The inner propeller shaft **6** extends in the front-rear direction, and a front portion of the inner propeller shaft **6** is disposed inside the outer propeller shaft **4**. The inner propeller shaft **6** is coaxial with the outer propeller shaft **4**. The inner propeller shaft **6** is supported to the outer propeller shaft **4** and the rear driven gear **49** by bearings **53**, **54** in a manner of being rotatable relative to the outer propeller shaft **4** and the rear driven gear **49**. The front driven gear **47** is splined (fitted) to a front end portion of the inner propeller shaft **6**. Accordingly, the front driven gear **47** and the inner propeller shaft **6** rotate integrally. The rear propeller **5** is attached to a rear end portion of the inner propeller shaft **6**.
(40) When the main drive gear **46** rotates integrally with the lower drive shaft **38**, the rotation is transmitted to the rear driven gear **49** and the front driven gear **47**. As a result, the outer propeller shaft **4** and the inner propeller shaft **6** rotate. At this time, rotation directions of the outer propeller shaft **4** and the inner propeller shaft **6** are in reverse. As the outer propeller shaft **4** and the inner propeller shaft **6** rotate, the front propeller **3** and the rear propeller **5** also rotate.
(41) When the positive rotation of the upper drive shaft **21** is transmitted to the outer propeller shaft **4** and the inner propeller shaft **6** by the clutch **27** through the intermediate drive shaft **35**, the lower drive shaft **38**, the lower gear mechanism **45**, and the like, a propulsive force that moves a ship forward is generated by the front propeller **3** and the rear propeller **5**. When the reverse rotation of the reverse output gear **25** is transmitted to the outer propeller shaft **4** and the inner propeller shaft **6** by the clutch **27** through the intermediate drive shaft **35**, the lower drive shaft **38**, the lower gear mechanism **45**, and the like, a propulsive force that moves the ship backward is generated by the front propeller **3** and the rear propeller **5**.
(42) (Setting Member Arrangement Hole and Gear Position Setting Member)
(43) FIG. **5** shows the lower gear chamber **15** in a state where the lower gear mechanism **45** and the like is not provided. As shown in FIG. **5**, the lower gear chamber **15** is formed with a setting member arrangement hole **61** in a front portion thereof in the lower case **11**. The setting member arrangement hole **61** is a bottomed hole that extends forward from the lower gear chamber **15** and has a circular cross-sectional shape. The setting member arrangement hole **61** is coaxial with the front driven gear **47** disposed in the lower gear chamber **15**. The setting member arrangement hole **61** is formed with, on a peripheral surface in the vicinity of an opening in a rear portion, a screw **62** having an axis extending in the front-rear direction. The screw **62** is a specific example of a “first screw”.
(44) As shown in FIG. **2**, the lower case **11** is formed with, above the setting member arrangement hole **61**, a vertical hole **63** extending downward from the upper portion of the lower case **11**. As shown in FIG. **5**, an insertion hole **65** is formed in a partition wall **64** between the vertical hole **63** and the setting member arrangement hole **61**. The insertion hole **65** penetrates the partition wall **64** in the upper-lower direction, and a lower end of the insertion hole **65** is opened at an uppermost portion of the peripheral surface of the setting member arrangement hole **61**. In addition, the insertion hole **65** is disposed at a front portion (portion on a back side) of the setting member arrangement hole **61**, and is located in front of the screw **62**. A stop hole **66** is formed in a lowermost portion of the peripheral surface of the setting member arrangement hole **61**. The stop hole **66** is a bottomed hole extending downward from the lowermost portion of the peripheral surface of the setting member arrangement hole **61**, and is located directly below the insertion hole **65** and is coaxial with the insertion hole **65**. In the present embodiment, the insertion hole **65** and the stop hole **66** have the same diameter.
(45) As shown in FIG. **4**, a gear position setting member **67** is provided in the setting member arrangement hole **61**. The gear position setting member **67** is located in front of the front driven gear **47**. The gear position setting member **67** has a function of setting the position of the front driven gear **47** in the front-rear direction.

(46) FIG. **6** shows the gear position setting member **67** as viewed from front. FIG. **7**A shows the gear position setting member **67** as viewed from left. FIG. **7**B shows a cross section of the gear position setting member **67** cut along a cutting line VII-VII in FIG. **6** as viewed from left, as well as a tool **81**.
(47) As shown in FIGS. **6** and **7**A, the gear position setting member **67** is made of, for example, a metal material, and has a bottomed cylindrical shape having an axis A extending in the front-rear direction. The gear position setting member **67** has an outer diameter set to a value slightly smaller than an inner diameter of the setting member arrangement hole **61** so that the gear position setting member **67** can rotate coaxially with the setting member arrangement hole **61** without being displaced in the upper-lower direction or in the left-right direction in the setting member arrangement hole **61**. The gear position setting member **67** is coaxial with the front driven gear **47**.
(48) A screw **68** having an axis extending in the front-rear direction is formed on an outer peripheral surface of the gear position setting member **67**. The screw **68** of the gear position setting member **67** is screwed to the screw **62** of the setting member arrangement hole **61**. The screw **68** is a specific example of a “second screw”.
(49) The screw **62** of the setting member arrangement hole **61** and the screw **68** of the gear position setting member **67** constitute a screw mechanism that moves the gear position setting member **67** in the front-rear direction in the setting member arrangement hole **61** when the gear position setting member **67** is rotated about the axis A. When the gear position setting member **67** disposed in the setting member arrangement hole **61** is viewed from rear and the gear position setting member **67** is rotated clockwise, the screw **68** rotates clockwise relative to the screw **62** so that the gear position setting member **67** moves forward. On the other hand, when the gear position setting member **67** is rotated counterclockwise, the screw **68** is rotated counterclockwise relative to the screw **62** so that the gear position setting member **67** is moved rearward.
(50) As shown in FIGS. **6** and **7**B, a tool attachment hole **69** for attaching the tool **81**, which rotates the gear position setting member **67** about the axis A, is provided in a central portion of the gear position setting member **67**. The tool attachment hole **69** is a hole whose opening shape is a square (for example, a hexagon). As shown in FIG. **8**C, the tool **81** includes, for example, a rod-shaped shaft portion **82** having a length for reaching the tool attachment hole **69** of the gear position setting member **67** disposed in the setting member arrangement hole **61** from rear of the lower case **11** and through the propeller shaft arrangement hole **17**. As shown in FIG. **7**B, the shaft portion **82** is provided with, at a tip end thereof, a fitting portion **83** having a rectangular (for example, hexagonal) cross section for being fittable into the tool attachment hole **69**. The tool attachment hole **69** is a specific example of a “tool attachment portion”.
(51) As shown in FIG. **4**, a rear end surface **67**A of the outer peripheral portion of the gear position setting member **67** is in contact with a front end surface of an outer ring **48**A of the bearing **48** that supports the front driven gear **47** to the lower case **11**. When the gear position setting member **67** is rotated clockwise using the tool **81**, the front driven gear **47** and the bearing **48** can be moved forward. When the gear position setting member **67** is rotated counterclockwise using the tool **81**, the front driven gear **47** and the bearing **48** can be pushed and moved rearward.
(52) As shown in FIG. **7**A, an annular portion **70** protruding forward is formed on the outer peripheral portion of the gear position setting member **67**, and a plurality of locking recesses **71** are formed on a front end of the annular portion **70** at equal intervals over the entire circumference. The number of the locking recesses **71** is an even number, for example, **16**. As shown in FIG. **4**, a fixing member **72** that non-rotatably fixes the gear position setting member **67** is attachably and detachably provided in the setting member arrangement hole **61** of the lower case **11**. The fixing member **72** is a long tubular member formed of, for example, a metal material. The fixing member **72** is inserted into the insertion hole **65** formed in the partition wall **64** and the stop hole **66** formed in the peripheral surface of the setting member arrangement hole **61**. The fixing member **72** has a diameter set to a value slightly smaller than the diameter of the insertion hole **65** and the stop hole

66 so that the fixing member **72** can be inserted into and removed from the insertion hole **65** and the stop hole **66** and is substantially exactly fitted into the insertion hole **65** and the stop hole **66**. When the fixing member **72** is inserted into the insertion hole **65** and the stop hole **66**, the fixing member **72** is engaged with, for example, two locking recesses **71** facing each other among the **16** locking recesses **71** of the gear position setting member **67** (see FIG. **6**). Accordingly, the gear position setting member **67** is non-rotatably fixed in the setting member arrangement hole **61**. Since the gear position setting member **67** is non-rotatably fixed by the fixing member **72**, positions of the front driven gear **47** and the bearing **48** in the front-rear direction can be determined.

(53) The fixing member **72** can be pulled out of the insertion hole **65** and the stop hole **66** by, for example, inserting a hand or a tool from the vertical hole **63**. When the fixing member **72** is pulled out, the gear position setting member **67** can be rotated using the tool **81**, and the positions of the front driven gear **47** and the bearing **48** in the front-rear direction can be changed by rotating the gear position setting member **67**.

(54) In the present embodiment, the fixing member **72** also functions as an oil supply pipe that supplies lubricating oil into the lower gear chamber **15**. As shown on lower left of FIG. **4**, the inside of the fixing member **72** is an oil flow portion **73** that flows the lubricating oil, and a lower end portion of the fixing member **72** is formed with an oil outflow hole **74** through which the lubricating oil flows out into the lower gear chamber **15**. An oil introduction pipe **75** through which the lubricating oil flows into the fixing member **72** from the outside of the lower case **11** can be connected to an upper end portion of the fixing member **72**.

(55) (Assembly of Lower Drive Shaft)

(56) FIGS. **8**A to **8**D show an operation process of assembling the lower drive shaft **38** integrally formed with the main drive gear **46** into the lower case **11**. In FIGS. **8**A to **8**D and **4**, P indicates a position where the front driven gear **47** meshes with the main drive gear **46** (specifically, a position in the front-rear direction of a rear end (tip end) of the front driven gear **47** in a state where the front driven gear **47** meshes with the main drive gear **46**). Hereinafter, this position is referred to as a “set position P”.

(57) In the outboard motor **1** of the present embodiment, the operation process of assembling the lower drive shaft **38** into the lower case **11** is as follows.

(58) First step: First, the operator rotates the gear position setting member **67** provided in the setting member arrangement hole **61** using the tool **81**, and moves the gear position setting member **67** to, for example, the vicinity of a foremost position in the setting member arrangement hole **61**. In this state, the operator inserts the bearing **48** and the front driven gear **47** into the lower gear chamber **15** from rear of the lower case **11** through the propeller shaft arrangement hole **17**, and attaches the bearing **48** and the front driven gear **47** to the front portion of the lower gear chamber **15**. Further, the operator pushes the bearing **48** and the front driven gear **47** forward so that the front end surface of the outer ring **48**A of the bearing **48** comes into contact with the rear end surface **67**A of the gear position setting member **67**. Accordingly, the front driven gear **47** is disposed in front of the set position P. In this way, since the gear position setting member **67** is moved to, for example, the vicinity of the foremost position in the setting member arrangement hole **61**, the front driven gear **47** can be disposed in front of the setting position P.

(59) Second step: as shown in FIG. **8**A, in a state where the front driven gear **47** is disposed in front of the set position P, the operator inserts the lower drive shaft **38** into the lower gear chamber **15** from rear of the lower case **11** through the propeller shaft insertion hole **65**. At this time, the operator inserts the lower drive shaft **38** into the propeller shaft arrangement hole **17** starting from an end portion (upper end portion) of the lower drive shaft **38** on a side where the main drive gear **46** is not integrally formed, and carries the lower drive shaft **38** into the lower gear chamber **15**.

(60) Third step: Next, the operator inserts the lower drive shaft **38** into the drive shaft arrangement hole **16** from below as shown in FIG. **8**B while rotating the lower drive shaft **38** inserted into the lower gear chamber **15** such that the end portion of the lower drive shaft **38** on the side where the

main drive gear **46** is not integrally formed faces upward. Then, the lower drive shaft **38** is attached to the drive shaft arrangement hole **16**.

(61) Fourth step: Next, as shown in FIG. **8**C, the operator inserts the tool **81** into the propeller shaft arrangement hole **17** from rear of the lower case **11**, further passes the tool **81** through the shaft hole **47**A of the front driven gear **47**, and fits the fitting portion **83** provided at the tip end of the tool **81** to the tool attachment hole **69** of the gear position setting member **67**.

(62) Fifth step: Next, the operator rotates the gear position setting member **67** and moves the gear position setting member **67** rearward by rotating the tool **81** in a direction of an arrow in FIG. **8**C. Accordingly, the bearing **48** and the front driven gear **47** are pushed by the gear position setting member **67** and move rearward. The operator rotates the tool **81**, moves the front driven gear **47** to the set position P, and meshes the front driven gear **47** with the main drive gear **46**.

(63) Sixth step: Next, as shown in FIG. **8**D, the operator inserts the fixing member **72** into the vertical hole **63** from above the lower case **11**, and sequentially inserts the fixing member **72** into the insertion hole **65**, the locking recesses **71** of the gear position setting member **67**, and the stop hole **66**. Accordingly, the gear position setting member **67** is non-rotatably fixed in the setting member arrangement hole **61**, the position of the gear position setting member **67** in the front-rear direction is fixed, and the position of the front driven gear **47** in the front-rear direction is set to the setting position P.

(64) At the time of manufacturing the outboard motor **1**, after the lower drive shaft **38** integrally formed with the main drive gear **46** and the front driven gear **47** are assembled into the lower case **11** in this way, for example, the operator assembles the rear driven gear **49**, the inner propeller shaft **6**, and the outer propeller shaft **4** into the lower gear chamber **15** and the propeller shaft arrangement hole **17**, couples the intermediate drive shaft **35** to the lower drive shaft **38** by the coupling member **36**, assembles the upper gear mechanism **22** and the clutch **27** into the upper gear chamber **14**, and the like.

(65) The second step and the third step are specific examples of a “drive shaft inserting step”. The fourth step is a specific example of a “tool attaching step”. The fifth step is a specific example of a “driven gear moving step”.

(66) (Adjustment of Mesh Between Main Drive Gear and Front Driven Gear)

(67) In the outboard motor **1** of the present embodiment, the operator can adjust the mesh (tooth contact, backlash, and the like) between the main drive gear **46** and the front driven gear **47** as follows. First, the operator detaches the rear driven gear **49**, the inner propeller shaft **6**, and the outer propeller shaft **4** from the lower case **11**. Next, the operator rotates the gear position setting member **67** using the tool **81** to change the position of the front driven gear **47** in the front-rear direction, thereby adjusting the mesh between the main drive gear **46** and the front driven gear **47**. Next, the operator assembles the rear driven gear **49**, the inner propeller shaft **6**, and the outer propeller shaft **4** to the lower case **11**.

(68) As described above, the outboard motor **1** of the present embodiment includes the gear position setting member **67** that sets the position of the front driven gear **47** in the front-rear direction, and the screw mechanism (screw **62**, screw **68**) that moves the gear position setting member **67** in the front-rear direction when the gear position setting member **67** is rotated about the axis A. Accordingly, by changing the position of the front driven gear **47** in the front-rear direction, the lower drive shaft **38** integrally formed with the main drive gear **46** can be assembled into the lower case **11**. That is, according to the outboard motor **1** of the present embodiment, by moving the gear position setting member **67** forward, the front driven gear **47** attached to the inside of the lower gear chamber **15** can be moved forward from the set position P. Accordingly, the front driven gear **47** can be moved away from the opening of the drive shaft arrangement hole **16** into the lower gear chamber **15**. As a result, in the lower gear chamber **15**, a vacant region below the drive shaft arrangement hole **16** can be expanded. Therefore, when the lower drive shaft **38** integrally formed with the main drive gear **46** is inserted into the drive shaft arrangement hole **16** from below through

the propeller shaft arrangement hole **17** and the lower gear chamber **15** from rear of the lower case **11**, it is possible to prevent the front driven gear **47** from coming into contact with the lower drive shaft **38** or the main drive gear **46** and from interfering with the insertion of the lower drive shaft **38** into the drive shaft arrangement hole **16**. Therefore, by the method for attaching the lower drive shaft **38** integrally formed with the main drive gear **46** to the drive shaft arrangement hole **16** by inserting the lower drive shaft **38** integrally formed with the main drive gear **46** into the drive shaft arrangement hole **16** from below through the propeller shaft arrangement hole **17** and the lower gear chamber **15** from rear of the lower case **11**, the lower drive shaft **38** integrally formed with the main drive gear **46** can be assembled into the lower case **11**.
(69) According to the outboard motor **1** of the present embodiment, since the lower drive shaft **38** integrally formed with the main drive gear **46** can be attached to the inside of the drive shaft arrangement hole **16** from below through the propeller shaft arrangement hole **17** and the lower gear chamber **15** of the lower case **11**, the lower drive shaft **38** integrally formed with the main drive gear **46** can be assembled into the lower case **11** without adding a large-diameter hole for passing the main drive gear **46** integrally formed with the lower drive shaft **38** to the lower case **11**. Therefore, it is possible to prevent the lateral width of the lower case **11** from increasing due to adding a large-diameter hole to the lower case **11**, and it is possible to prevent the resistance of water from increasing during navigation.
(70) According to the outboard motor **1** of the present embodiment, since the gear position setting member **67** and the screw mechanism (screws **62**, **68**) are provided, it is possible to easily and quickly adjust the mesh between the main drive gear **46** and the front driven gear **47** by changing the position of the front driven gear **47** in the front-rear direction. That is, in the outboard motor in the related art, the mesh between the drive gear and the front driven gear is adjusted by a method of replacing a shim attached to a boss of the front driven gear with another shim having a different thickness. For this reason, to adjust the mesh between the drive gear and the front driven gear, it is necessary to detach the drive gear and the front driven gear from the lower case. In contrast, according to the outboard motor **1** of the present embodiment, the mesh between the main drive gear **46** and the front driven gear **47** can be adjusted by changing the position of the front driven gear **47** in the front-rear direction in a state where the main drive gear **46** and the front driven gear **47** are attached to the lower case **11**. Therefore, when the mesh between the main drive gear **46** and the front driven gear **47** is adjusted, the step of attaching and detaching the main drive gear **46** and the front driven gear **47** to and from the lower case **11** can be omitted. Therefore, it is possible to reduce the number of operation steps for adjusting the mesh between the main drive gear **46** and the front driven gear **47**, and it is possible to facilitate and speed up the operation for adjusting the mesh. According to the present embodiment, such an operational effect can be obtained by a simple structure in which the gear position setting member **67** is moved by the screw mechanism.
(71) In the outboard motor **1** of the present embodiment, the tool attachment hole **69** for attaching the tool **81** that rotates the gear position setting member **67** is provided in the central portion of the gear position setting member **67**. With this configuration, the tool **81** is attached to the gear position setting member **67** from rear of the lower case **11** through the propeller shaft arrangement hole **17** and the shaft hole **47**A of the front driven gear **47**, and the gear position setting member **67** is rotated using the tool **81**, so that the position of the front driven gear **47** in the front-rear direction can be easily changed. Therefore, the lower drive shaft **38** integrally formed with the main drive gear **46** can be easily assembled into the lower case **11**, and the mesh between the main drive gear **46** and the front driven gear **47** can be easily adjusted.
(72) The outboard motor **1** of the present embodiment includes the fixing member **72** that is attachably and detachably provided in the setting member arrangement hole **61** formed in the lower gear chamber **15** and that non-rotatably fixes the gear position setting member **67**. With this configuration, the operator can switch between a state in which the gear position setting member **67** is movable in the front-rear direction and a state in which the gear position setting member **67** is

not movable in the front-rear direction simply by attaching and detaching the fixing member **72** to and from the setting member arrangement hole **61**. Therefore, the lower drive shaft **38** integrally formed with the main drive gear **46** can be more easily assembled into the lower case **11**, and the mesh between the main drive gear **46** and the front driven gear **47** can be more easily adjusted. In particular, according to the present embodiment, since the gear position setting member **67** can be switched between the state in which the gear position setting member **67** is movable in the front-rear direction and the state in which the gear position setting member **67** is not movable in the front-rear direction simply by inserting and removing the fixing member **72** into and from the insertion hole **65**, the stop hole **66**, and the locking recesses **71** of the gear position setting member **67**, the assembly of the lower drive shaft **38** and the mesh adjustment can be performed more easily.

(73) In the outboard motor **1** of the present preferred embodiment, the intermediate drive shaft **35** and the lower drive shaft **38** are coupled by the coupling member **36**. With this configuration, it is possible to reduce the length of the lower drive shaft **38** integrally formed with the main drive gear **46**. Therefore, the lower drive shaft **38** integrally formed with the main drive gear **46** can be smoothly inserted into the drive shaft arrangement hole **16** from below after passing through the propeller shaft arrangement hole **17** and the lower gear chamber **15** from rear of the lower case **11**.

(74) In the present embodiment, the fixing member **72** has a function of an oil supply pipe that supplies the lubricating oil into the lower gear chamber **15**, in addition to a function of fixing the gear position setting member **67**. Accordingly, the number of oil supply pipes can be reduced, and the number of components of the outboard motor **1** can be reduced.

(75) In the present embodiment, since the main drive gear **46** is formed integrally with the lower drive shaft, the coupling strength between the lower drive shaft **38** and the main drive gear **46** can be increased. Therefore, the stability of power transmission from the lower drive shaft **38** to the propeller shafts **4**, **6** can be enhanced.

(76) The screw mechanism that moves the gear position setting member in the front-rear direction when the gear position setting member is rotated is not limited to the screws **62**, **68** of the above-described embodiment. The tool attachment portion of the gear position setting member is not limited to the tool attachment hole **69** of the above-described embodiment. The configuration of a locking portion that engages with the fixing member and disables the rotation of the gear position setting member is not limited to the locking recesses **71** of the above-described embodiment. As shown in FIGS. **9**A to **9**C, the screw mechanism may be configured such that a protruding portion **92** protruding rearward from the center of a setting member arrangement hole **91** is provided on a front surface (bottom surface) of the setting member arrangement hole **91**, a screw **93** is formed on an outer peripheral surface of the protruding portion **92**, a screw hole **95** formed with a screw **96** on an inner peripheral surface is formed in a center portion of a front portion (bottom portion) of a gear position setting member **94** formed in a bottomed cylindrical shape, the protruding portion **92** is inserted into the screw hole **95**, and the screw **96** and the screw **93** are screwed together. The tool attachment portion may be configured such that a tool attachment protrusion **97**, which protrudes rearward from the central portion of the gear position setting member **94** and has a rectangular (for example, hexagonal) cross section, is provided on the front portion (bottom portion) of the gear position setting member **94**. The locking portion may be configured such that locking holes **98** that are elongated in the front-rear direction are provided in a peripheral wall of the gear position setting member **94**.

(77) In the above embodiment, the tubular fixing member **72** having the function of an oil supply pipe is described as an example, and the fixing member may be a simple pin-shaped or rod-shaped member not having the function of an oil supply pipe.

(78) The present disclosure is not limited to a contra-rotating propeller outboard motor. The present disclosure can also be applied to an outboard motor including one propeller shaft as long as the outboard motor includes a drive gear coupled to a lower end side of a drive shaft and a driven gear

meshing with the drive gear from front of the drive gear.
(79) The present disclosure can be appropriately changed without departing from the gist or idea of the disclosure that can be read from the claims and the entire specification, and an outboard motor and a drive shaft assembling method accompanied by such a change are also included in the technical idea of the present disclosure.

Claims

1. An outboard motor comprising: a case having a gear chamber, a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber; a drive shaft disposed in the drive shaft arrangement hole and extending in an upper-lower direction; a propeller shaft disposed in the propeller shaft arrangement hole and extending in a front-rear direction; a drive gear disposed in the gear chamber and coupled to a lower end side of the drive shaft, the drive gear being a bevel gear; a driven gear disposed in front of the drive gear in the gear chamber, coupled to a front end side of the propeller shaft, and configured to mesh with the drive gear, the driven gear being a bevel gear; a gear position setting member provided in front of the driven gear in the case and configured to set a position of the driven gear in the front-rear direction; and a screw mechanism configured to move the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member, the axis of the gear position setting member extending in the front-rear direction, wherein the screw mechanism includes a first screw provided in the case and having an axis extending in the front-rear direction, and a second screw provided on the gear position setting member, having an axis extending in the front-rear direction, and configured to be screwed with the first screw.
2. The outboard motor according to claim 1, wherein the gear position setting member is formed in a columnar shape or a bottomed cylindrical shape having an axis extending in the front-rear direction, and wherein the gear position setting member is provided with, a tool attachment portion to which a tool that rotates the gear position setting member is attached, at a central portion of the gear position setting member.
3. The outboard motor according to claim 1, further comprising: a fixing member attachably and detachably provided in the case and configured to non-rotatably fix the gear position setting member to the case.
4. The outboard motor according to claim 1, wherein the drive gear is formed integrally with the drive shaft.
5. A drive shaft assembling method for an outboard motor, the outboard motor including: a case having a gear chamber, a drive shaft arrangement hole communicating with the gear chamber from above the gear chamber, and a propeller shaft arrangement hole communicating with the gear chamber from rear of the gear chamber; a drive shaft disposed in the drive shaft arrangement hole and extending in an upper-lower direction; a propeller shaft disposed in the propeller shaft arrangement hole and extending in a front-rear direction; a drive gear disposed in the gear chamber and integrally formed with a lower end side of the drive shaft, the drive gear being a bevel gear; a driven gear disposed in front of the drive gear in the gear chamber, coupled to a front end side of the propeller shaft, and configured to mesh with the drive gear, the driven gear being a bevel gear; a gear position setting member provided in front of the driven gear in the case and configured to set a position of the driven gear in the front-rear direction; and a screw mechanism configured to move the gear position setting member in the front-rear direction when the gear position setting member is rotated about an axis of the gear position setting member that extends in the front-rear direction, the drive shaft assembling method for assembling the drive shaft integrally formed with the drive gear into the case, and the drive shaft assembling method comprising: inserting the drive shaft into the drive shaft insertion hole from the gear chamber after inserting the drive shaft into the gear

chamber through the propeller shaft insertion hole in a state where the driven gear is disposed in front of a set position where the driven gear is meshed with the drive gear; inserting a tool that rotates the gear position setting member into the gear chamber through the propeller shaft insertion hole and attaching the tool to the gear position setting member; and rotating the gear position setting member by the tool to move the driven gear to the set position to mesh with the drive gear.